

# RoundFi — Stress Lab and Economic Model (Updated Professional Edition)

**Document 05 — Updated after PRs 400–405, ECO-001..007, SEV-047/048 and regression-guard hardening.**

Economic stress testing, bounded solvency modeling, behavioral-credit assumptions and operational risk framing for RoundFi on Solana.

## 1. Executive Summary

RoundFi is a behavioral-credit protocol built on Solana. Its central idea is not simply rotating capital or creating another DeFi lending primitive. The core thesis is that recurring financial behavior may become a reusable trust signal over time.

The Stress Lab exists to test whether the economic architecture supporting this thesis remains coherent under modeled stress conditions. It is a simulation, verification and adversarial testing environment designed to expose weaknesses before progressive mainnet exposure.

The purpose of this document is not to claim impossible failure resistance or permanent solvency guarantees. The correct framing is narrower and more credible: RoundFi remained solvent within currently modeled and tested scenarios, while known economic risks are progressively transformed into regression protections.

The protocol still faces meaningful unknowns. Human behavior, coordination durability, default clustering and real-world adoption remain partially unvalidated at scale. The Stress Lab should therefore be interpreted as an evolving economic verification environment rather than a claim of final certainty.

## 2. Why the Stress Lab Exists

RoundFi combines multiple economic and behavioral layers simultaneously: recurring contributions, rotating payouts, reputation-based collateral reduction, adaptive escrow, solidarity reserves, yield routing and default handling. This creates a system more complex than traditional overcollateralized lending.

Because the protocol introduces dynamic behavioral assumptions, deterministic testing becomes essential. The Stress Lab was created to answer questions that cannot be resolved through narrative alone.

Key questions include: What happens when a member defaults after receiving early capital access? What happens when multiple defaults occur together? What happens under zero-yield conditions? What happens if behavioral assumptions weaken over time?

The Stress Lab therefore acts as a bounded simulation environment where the team can continuously challenge solvency assumptions, accounting consistency and failure containment before aggressive scale expansion.

## 3. What the Stress Lab Is

The Stress Lab is currently implemented as a pure TypeScript economic simulator combined with an interactive front-end route. It models pool contributions, escrow balances, stake behavior, default events, yield assumptions, reserve routing and solvency outcomes.

The `/lab` route allows reviewers and contributors to interact directly with predefined economic scenarios. The simulator acts both as an explanatory environment and as a parity reference against on-chain behavior.

Recent operational hardening introduced a stronger focus on regression consistency. Previously identified failure paths are increasingly converted into permanent regression checks rather than remaining purely theoretical discussions.

This transition matters because economic systems degrade silently if edge cases are not replayed consistently over time. The Stress Lab is therefore evolving from a conceptual simulator into an institutional-grade regression and adversarial testing environment.

## 4. Core Behavioral Thesis

The central hypothesis of RoundFi remains unchanged: users who repeatedly complete financial commitments over time may become lower-risk participants than wallets with no behavioral history.

Traditional DeFi systems primarily ask: “How much collateral can this wallet lock?” RoundFi asks a different question: “Has this wallet demonstrated recurring financial discipline across time?”

This distinction is important because the protocol is not attempting to eliminate risk through collateral alone. Instead, it attempts to combine stake, escrow, reserves, recurring obligations and behavioral history into a broader underwriting framework.

The Stress Lab exists to test whether this broader framework remains coherent when exposed to adversarial conditions and coordination stress.

## 5. Pool Structure and Participation Model

A RoundFi pool is a structured financial coordination group with recurring installments and scheduled payout behavior. Members contribute over time and receive access to capital according to pool rules.

Typical pools include fixed member counts, recurring installment values, stake requirements, payout ordering, solidarity reserves and escrow logic. Reputation level influences collateral requirements and therefore affects capital efficiency.

The economic architecture attempts to create alignment between recurring participation and future access to capital. This is fundamentally different from static collateral-only systems.

The current implementation uses smaller pools in Devnet and canary-stage testing environments. However, the economic logic is designed to scale progressively as validation increases.

## 6. Reputation Ladder and Collateral Reduction

The protocol currently models a simplified behavioral ladder:

- Level 1 — 50% stake requirement
- Level 2 — 30% stake requirement
- Level 3 — 10% stake requirement

This ladder should not be interpreted as free leverage or unsecured borrowing. Users still maintain future obligations, escrow exposure and participation commitments.

The 10% tier remains the most economically sensitive scenario because it introduces the largest gap between posted stake and accessible capital. For this reason, veteran-default scenarios receive disproportionate attention in the Stress Lab.

Importantly, RoundFi should avoid claiming that behavioral reputation has already proven capable of replacing collateral at scale. That hypothesis remains under empirical validation.

## 7. Triple Shield Architecture (Updated Framing)

The Triple Shield is best understood as a layered mitigation architecture rather than a guarantee mechanism. The goal is to reduce modeled loss exposure through multiple overlapping defenses.

Shield 1 introduces structurally conservative early-cycle behavior through Seed Draw retention. Instead of immediately draining pool liquidity during initial cycles, the architecture retains a significant portion of capital inside the system.

Shield 2 combines adaptive escrow and member stake. Escrow release may remain conditional on continued participation, while stake creates direct economic exposure for defaulting participants.

Shield 3 combines solidarity reserves, deterministic accounting and yield routing. Yield is treated as a supporting optimization layer rather than the foundation of solvency.

## 8. Default Handling and Loss Bounding

Default handling is one of the most important areas of the protocol because it directly tests whether the behavioral-credit thesis can survive adverse conditions.

The current architecture attempts to bound losses through multiple paths: solidarity reserves, escrow seizure, stake seizure and deterministic accounting logic.

The protocol should avoid presenting defaults as impossible. Defaults are expected to occur eventually in any meaningful financial coordination system. The objective is not zero defaults; the objective is controlled and bounded loss behavior.

This distinction is important institutionally because exaggerated solvency claims damage credibility. The updated posture of the Stress Lab intentionally favors bounded, testable and falsifiable claims over deterministic promises.

## 9. Stress Scenarios

The Stress Lab includes multiple scenario categories designed to progressively challenge protocol assumptions.

Healthy scenarios model normal recurring behavior with orderly payments, predictable payouts and positive reputation progression.

Pre-default scenarios test stress detection before final settlement events occur. Post-default scenarios test whether accounting remains coherent after seizure and settlement execution.

Cascade scenarios model sequential stress events and attempt to identify whether one failure path introduces secondary accounting inconsistencies. Triple Veteran Default scenarios specifically challenge the lowest-collateral tiers under concentrated stress conditions.

## 10. Zero-Yield and Adversarial Assumptions

RoundFi should not depend on optimistic yield assumptions to remain economically coherent. The protocol therefore models low-yield, unstable-yield and zero-yield conditions.

Yield is intentionally framed as an efficiency enhancer rather than the source of solvency. The protocol's primary protections remain stake, escrow, reserves and bounded accounting logic.

This conservative posture became significantly more important after operational hardening and judge-facing review cleanup. The protocol now intentionally avoids any framing that could imply yield guarantees or deterministic profitability.

Adversarial assumptions are now treated as a core component of the economic model rather than an optional consideration.

## 11. Regression Guards and Economic Hardening

PRs 404–405 introduced a major conceptual improvement: known economic risks are progressively transformed into permanent regression protections.

This includes ECO-001..007 hardening, SEV-047/048 resolution, insolvency reachability validation and replay testing against previously identified edge cases.

The goal is not to claim that the protocol can never fail. The goal is to reduce the probability that previously known failure paths silently reappear during future development.

This transition meaningfully increased the maturity of the protocol because economic safety is now treated as a continuously monitored engineering problem rather than a static theoretical document.

## 12. Human Coordination Risk

The largest unresolved variable remains human behavior. The protocol still lacks production-scale evidence regarding how users behave under meaningful financial pressure.

Critical unanswered questions include: How sensitive are users to reputation changes? Does social pressure reduce default frequency? Do participants continue behaving cooperatively when economic stress increases?

These uncertainties matter because RoundFi ultimately depends on coordination quality rather than collateral alone.

The protocol therefore requires progressive real-world experimentation, behavioral telemetry and empirical validation before stronger economic conclusions can be justified.

## 13. Current Validation Status

The current repository status includes live front-end demonstrations, Devnet deployment, Stress Lab simulation, on-chain settlement flows, browser-signed contribution flows and default-settlement scaffolding.

The project also demonstrates operational hardening uncommon for hackathon-stage Solana infrastructure, including canary discipline, freeze governance, regression awareness and adversarial replay thinking.

However, Devnet success should not be interpreted as equivalent to production-grade validation. External audit, real-world behavioral data and progressive operational exposure are still required.

## 14. Risks and Limitations

Model risk remains significant because simulated assumptions may diverge from real-world participant behavior.

Implementation risk remains important because on-chain behavior must continue matching the economic simulator and regression assumptions.

Default clustering risk remains unresolved because real-world stress events may exceed current modeled assumptions.

Adoption risk also remains meaningful. Even if the mathematics remain coherent, the protocol still requires sustained user participation and ecosystem demand.

## 15. Strategic Interpretation

The Stress Lab should be interpreted as a bounded economic verification environment rather than a proof of impossible failure.

RoundFi is strongest when presented as an evolving behavioral-credit infrastructure system under rigorous testing and empirical refinement.

Institutional credibility increases when the protocol clearly distinguishes between:

- what has been modeled;
- what has been tested;
- what remains uncertain;
- what still requires real-world validation.

## 16. Final Positioning

RoundFi is evolving toward programmable behavioral trust infrastructure rather than a purely collateral-driven DeFi primitive.

The protocol's long-term value depends less on optimistic APY assumptions and more on whether behavioral reputation can eventually reduce financial friction in measurable ways.

The Stress Lab matters because it transforms that thesis into something testable, falsifiable and progressively verifiable.

## 17. Final Audit Snapshot

| Area                  | Current Assessment                      |
|-----------------------|-----------------------------------------|
| Economic Hardening    | Strongly improved after ECO-001..007    |
| Behavioral Validation | Still early-stage and unproven at scale |
| Operational Maturity  | High for current protocol stage         |
| Main Unknown          | Human coordination under real pressure  |
| Overengineering Risk  | Moderate and increasing                 |
| Strategic Strength    | Behavior-first trust architecture       |

## 18. Related Documents

- 01-roundfi-overview.md
- 02-technical-whitepaper.md
- 03-architecture-spec.md
- 04-behavioral-reputation-score.md
- 06-market-and-gtm.md
- 07-business-model-b2b-oracle.md
- 09-risk-and-compliance.md
- 11-devnet-status-and-proof.md